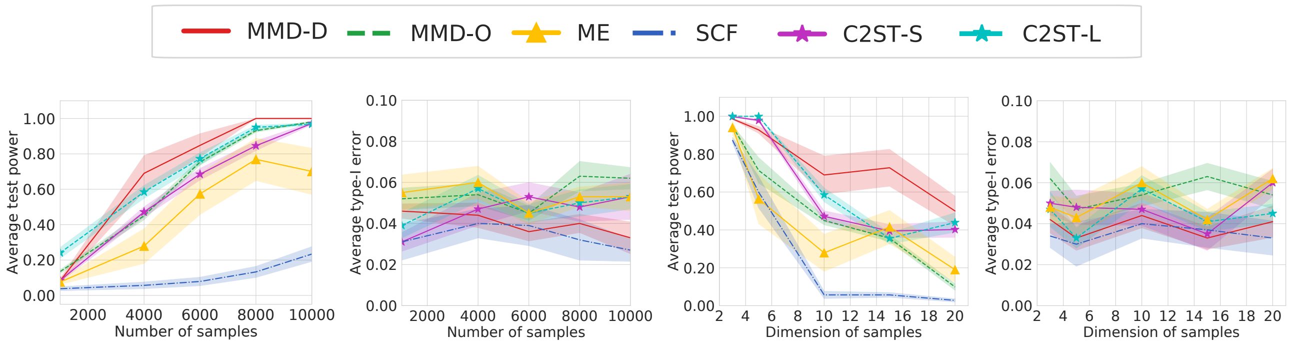

(b) Power vs. N; $d = 10$ (c) Level vs. N; $d = 10$ (d) Power vs. d; $N = 4\,000$ (e) Level vs. d; $N = 4\,000$

Figure 3. Results on *HDGM-S* and *HDGM-D* for $\alpha = 0.05$ (black line). Left: average test power (a) and Type I error (b) when increasing the number of samples N, keeping $d = 10$. Right: average test power (c) and Type I error (d) when increasing the dimension d, keeping $N = 4\,000$. Shaded regions show standard errors for the mean.

- MMD-D: MMD with a deep kernel; our method described in Section 5.
- MMD-O: MMD with a Gaussian kernel whose lengthscale is optimized as in Section 5. This gives better results than standard heuristics.
- Mean embedding (ME): a state-of-the-art test (Chwialkowski et al., 2015; Jitkrittum et al., 2016) based on differences in Gaussian kernel mean embeddings at a set of optimized points.
- Smooth characteristic functions (SCF): a state-of-the-art test (Chwialkowski et al., 2015; Jitkrittum et al., 2016) based on differences in Gaussian mean embeddings at a set of optimized frequencies.
- Classifier two-sample tests, including C2STS-S (Lopez-Paz & Oquab, 2017) and C2ST-L (Chen & Cloninger, 2019) as described in Section 4. We set the test thresholds via permutation for both.

For synthetic datasets, we take a single sample set for $S_{\mathbb{P}}^{tr}$ and $S_{\mathbb{Q}}^{tr}$ and learn a kernel/test locations/etc once for each method on that training set. We then evaluate its rejection rate on 100 new sample sets $S_{\mathbb{P}}^{te}$, $S_{\mathbb{Q}}^{te}$ from the same distribution. For real datasets, we select a subset of the available data for $S_{\mathbb{P}}^{tr}$ and $S_{\mathbb{Q}}^{tr}$ and train on that; we then evaluate on 100 random subsets, disjoint from the training set, of the remaining data. We repeat this full process 10 times, and report the mean rejection rate of each test. Table 5 shows significance tests. Further details are in Appendix B.

***Blob* dataset.** *Blob-D* is the dataset shown in Figure 1; *Blob-S* has $\mathbb{Q}$ also equal to the distribution shown in Figure 1a, so that the null hypothesis holds. Details are given in Table 6 (Appendix B.1).

Results are shown in Figure 2. MMD-D and C2ST-L are the clear winners in power, with MMD-D better in the higher-sample regime, and MMD-D is more reliable than C2STs. Figure 2c shows that J is higher for MMD-D than MMD-O, in addition to the actual test power being better, as discussed in Section 3. All methods have expected Type I error rates.

High-dimensional Gaussian mixtures. Here we study bimodal Gaussian mixtures in increasing dimension. Each distribution has two Gaussian components; in *HDGM-S*, $\mathbb{P}$ and $\mathbb{Q}$ are the same, while in *HDGM-D*, $\mathbb{P}$ and $\mathbb{Q}$ differ in the covariance of a single dimension pair but are otherwise the same. Details are in Table 6 (Appendix B.1). We consider both increasing N while keeping $d = 10$ and increasing d while keeping $N = 4\,000$, with results shown in Figure 3. Again, MMD-D has generally the best test power across a range of problem settings, with reasonable type I error.

***Higgs* dataset (Baldi et al., 2014).** We compare the jet ϕ-momenta distribution ($d = 4$) of the background process, $\mathbb{P}$, which lacks Higgs bosons, to the corresponding distribution $\mathbb{Q}$ for the process that produces Higgs bosons, following Chwialkowski et al. (2015). As discussed in these previous works, ϕ-momenta carry very little discriminating information for recognizing whether Higgs bosons were produced. We consider a series of tests with increased number of samples N.

We report average test power (comparing $\mathbb{P}$ to $\mathbb{Q}$) in Table 1, and average type-I error (comparing $\mathbb{P}$ to $\mathbb{P}$ or $\mathbb{Q}$ to $\mathbb{Q}$) in Table 7 (Appendix B.6). As before, MMD-D generally performs the best; although the improvement over MMD-O here is not dramatic, MMD-D does notably outperform C2ST. All methods maintain reasonable Type I errors.

***MNIST* generative model.** The *MNIST* dataset contains 70 000 handwritten digit images (LeCun et al., 1998). We compare true *MNIST* data samples $\mathbb{P}$ to samples $\mathbb{Q}$ from a pretrained deep convolutional generative adversarial network (DCGAN) (Radford et al., 2016). Samples from both distributions are shown in Figure 4 (in Appendix B.2).

We consider tests for increasing numbers of samples N, and report average test power (for $\mathbb{P}$ to $\mathbb{Q}$) in Table 2 and

Table 1. Higgs ($\alpha = 0.05$): average test power±standard error for N samples. Bold represents the highest mean per row.

N	ME	SCF	C2ST-S	C2ST-L	MMD-O	MMD-D
1 000	0.120±0.007	0.095±0.022	0.082±0.015	0.097±0.014	**0.132±0.005**	0.113±0.013
2 000	0.165±0.019	0.130±0.026	0.183±0.032	0.232±0.017	0.291±0.012	**0.304±0.035**
3 000	0.197±0.012	0.142±0.025	0.257±0.049	0.399±0.058	0.376±0.022	**0.403±0.050**
5 000	0.410±0.041	0.261±0.044	0.592±0.037	0.447±0.045	0.659±0.018	**0.699±0.047**
8 000	0.691±0.067	0.467±0.038	0.892±0.029	0.878±0.020	0.923±0.013	**0.952±0.024**
10 000	0.786±0.041	0.603±0.066	0.974±0.007	0.985±0.005	**1.000±0.000**	**1.000±0.000**
Avg.	0.395	0.283	0.497	0.506	0.564	**0.579**

Table 2. MNIST ($\alpha = 0.05$): average test power±standard error for comparing N real images to N DCGAN samples.

N	ME	SCF	C2ST-S	C2ST-L	MMD-O	MMD-D
200	0.414±0.050	0.107±0.018	0.193±0.037	0.234±0.031	0.188±0.010	**0.555±0.044**
400	0.921±0.032	0.152±0.021	0.646±0.039	0.706±0.047	0.363±0.017	**0.996±0.004**
600	**1.000±0.000**	0.294±0.008	**1.000±0.000**	0.977±0.012	0.619±0.021	**1.000±0.000**
800	**1.000±0.000**	0.317±0.017	**1.000±0.000**	**1.000±0.000**	0.797±0.015	**1.000±0.000**
1 000	**1.000±0.000**	0.346±0.019	**1.000±0.000**	**1.000±0.000**	0.894±0.016	**1.000±0.000**
Avg.	0.867	0.243	0.768	0.783	0.572	**0.910**

average Type I error ($\mathbb{P}$ to $\mathbb{P}$) in Table 8 (in Appendix B.6). MMD-D substantially outperforms its competitors in test power, with the desired Type I error. ME also does well in this case: it is perhaps particularly suited to this problem, since it is capable of identifying either modes dropped by the generative model or spurious modes it inserts.

***CIFAR-10* vs *CIFAR-10.1*.** *CIFAR-10.1* (Recht et al., 2019) is an attempt to collect a new test set for the very popular *CIFAR-10* image classification dataset (Krizhevsky, 2009). Normally, when evaluating a supervised model, we consider the test set an independent sample from the training distribution, ideally never-before-seen by the training algorithm. But modern computer vision model architectures and training procedures have been developed based on repeatedly evaluating on the *CIFAR-10* test set ($\mathbb{P}$), so it is possible that current models themselves are dependent on $\mathbb{P}$. *CIFAR-10.1* ($\mathbb{Q}$) is an attempt at an independent sample from this distribution, collected after the models were trained, so that they are truly independent of $\mathbb{Q}$. These models do obtain substantially lower accuracies on $\mathbb{Q}$ than on $\mathbb{P}$ – but this drop is surprisingly consistent across models, which seems unlikely to be due to the expected overfitting. The main potential explanation proposed by Recht et al. is dataset shift, but their attempt (in their Appendix C.2.8) at what amounts to a C2ST-S did not reject $\mathfrak{H}_0$.[4] Samples from each distribution are shown in Figure 5 (Appendix B.2).

We train on 1 000 images from each dataset and test on 1 031, so that we use the entirety of *CIFAR-10.1* each time, and average over ten repetitions. These tests provide strong evidence (Table 3) that images in the *CIFAR-10.1* test set *are* statistically different from the *CIFAR-10* test set, with MMD-D again strongest and ME still performing well.

Table 3. CIFAR-10.1 ($\alpha = 0.05$): mean rejection rates.

ME	SCF	C2ST-S	C2ST-L	MMD-O	MMD-D
0.588	0.171	0.452	0.529	0.316	**0.744**

Our learned kernel also helps provide some ability to interpret the difference between $\mathbb{P}$ and $\mathbb{Q}$, particularly if we use it for an ME test. Appendix C explores this.

Recht et al. (2019) also provide a new ImageNetV2 test set for the ImageNet dataset, with similar properties; we defer this more challenging problem to future work.

7.2. Ablation Study

We now study in more detail the difference between MMD-D and closely related methods. Recall from Section 4 that there are two main differences between MMD-D and C2STs: first, using a "full" kernel (1) rather than the sign-based kernel (6) or the intermediate linear kernel (7). Second, training to maximize $\hat{J}_\lambda$ (4) rather than a cross-entry surrogate. MMD-D uses a full kernel (1) trained for test power; C2ST-S effectively uses the sign kernel (6) trained for cross entropy.

In this section, we consider the performance of several intermediate models empirically, demonstrating that both factors help in testing. All are based on the same feature extraction architecture ϕ_ω; some models add a classification layer with new parameters w and b,

$$f_\omega(x) = w^\mathsf{T}\phi_\omega(x) + b,$$

[4] Assuming pretrained classifiers are independent of $\mathbb{P}$, Figure 1 of Recht et al. (2019) indicates that the joint (images, labels) distribution certainly differs between *CIFAR-10* and *CIFAR-10.1*. We test here whether the marginal image distribution differs.

Table 4. Mean test power on *Blob* ($n_b = 40$), *HDGM* ($N = 4000, d = 10$), *Higgs* ($N = 3000$) and *MNIST* ($N = 400$) for $\alpha = 0.05$. See Section 7.2 for the naming scheme; S+C corresponds to C2ST-S, L+C to C2ST-L, and D+J to MMD-D. L+M is the method proposed by Kirchler et al. (2020).

	S+C	L+C	G+C	D+C	L+M	G+M	D+M	L+J	G+J	D+J
Blob	0.835	0.942	0.901	0.900	0.851	0.960	0.906	0.952	0.966	**0.985**
HDGM	0.472	0.585	0.287	0.302	0.494	0.223	0.539	0.635	0.604	**0.659**
Higgs	0.257	0.399	0.353	0.384	0.321	0.254	0.379	0.295	0.364	**0.403**
MNIST	0.646	0.706	0.784	0.803	0.845	0.680	0.760	0.935	0.976	**0.996**
Avg.	0.553	0.658	0.581	0.597	0.628	0.529	0.646	0.704	0.727	**0.761**

Table 5. Paired t-test results ($\alpha = 0.05$) for the results of Section 7.1. For *HDGM*, we fix $d = 10$ (corresponding to Figure 3a). ✓ indicates MMD-D achieved statistically significantly higher mean test power than the other method, × that it did not.

Dataset	ME	SCF	C2ST-S	C2ST-L	MMD-O
Blob	✓	✓	✓	×	×
HDGM	✓	✓	✓	✓	✓
Higgs	✓	✓	✓	×	×
MNIST	✓	✓	✓	✓	✓

which is treated as outputting classification logits. The model variants we consider are

S A kernel $\mathbb{1}(f_\omega(x) > 0)\mathbb{1}(f_\omega(y) > 0)$; corresponds to a test statistic of the accuracy of f (Proposition 3).

L A kernel $f_\omega(x)f_\omega(y)$; corresponds to a test statistic comparing the mean value of f (Proposition 4).

G A Gaussian kernel $\kappa(\phi_\omega(x), \phi_\omega(y))$.

D The deep kernel (1) based on ϕ_ω.

We combine these model variants with a suffix describing the optimization objective:

J Choose ω, including possibly w and b, to optimize the approximate test power (4).

M Choose ω, including possibly w and b, to maximize the value of the empirical MMD between two samples.[5]

C Choose ω, including w and b, to optimize cross-entropy using the classifier that specifies the probability of x belonging to $\mathbb{P}$ as $1/\left(1 + \exp(-f_\omega(x))\right)$.[6]

Table 4 presents results for all of these methods (except for S+J, which is non-differentiable and hence difficult to optimize). Performance generally improves as we move from S to L to G to D, and from C to J.

[5] If a deep kernel is unbounded, directly maximizing MMD will make optimized parameters of ϕ_ω be infinite. Thus, for L+M, we consider a normalized linear deep kernel: $\tanh(f_\omega(x)/\|S\|_{\mathrm{F}})\tanh(f_\omega(y)/\|S\|_{\mathrm{F}})$, where $S = [S_{\mathbb{P}}; S_{\mathbb{Q}}]$ and $\|\cdot\|_F$ is the Frobenius norm.

[6] G+C and D+C take the fixed ϕ_ω embeddings, then find the optimal lengthscale/etc by optimizing $\hat{J}_\lambda$.

7.3. Architecture design of deep kernels

For *Blob*, *HDGM* and *Higgs*, ϕ_ω is a five-layer fully-connected neural network, with softplus activations. the number of neurons in hidden and output layers of ϕ_ω are set to 50 for *Blob*, $3d$ for *HDGM* and 20 for *Higgs*, where d is the dimension of samples. in general, we expect similar fully-connected networks, to be reasonable choices for datasets where strong structural assumptions are not known, perhaps with $3d$ as a baseline width for datasets of at least moderate dimension.

For *MNIST* and *CIFAR*, ϕ_ω is a *convolutional neural network* (CNN) that contains four convolutional layers and one fully-connected layer. The structure of the CNN follows the structure of the feature extractor in the DCGAN's discriminator (Radford et al., 2016) (see Figures 6 and 8 for the structure of ϕ_ω in MMD-D, and Figures 7 and 9 for the structure of classifier F in C2ST-S and C2ST-L). In general, we expect GAN discriminator architectures to work well for image datasets, as the problem is closely related.

8. Conclusions

The test power of MMD is limited by simple kernels (e.g., Gaussian kernel or other translation-invariant kernels) when facing complex-structured distributions, but we can avoid this problem with richer *deep kernels*, which is no longer translation-invariant. We show that optimizing the parameters of these kernels to maximize the test power, as proposed by Sutherland et al. (2017), outperforms state-of-the-art alternatives even when considering large, deep kernels with hundreds of thousands of parameters, rather than the simple shallow kernels they considered. We provide theoretical guarantees that this process is reasonable to conduct on finite samples, and asymptotically selects the most powerful kernel. We also give deeper insight into the relationship between this approach and classifier two-sample tests (Lopez-Paz & Oquab, 2017), explaining why this approach outperforms that one.

We thus recommend practitioners to use optimized deep kernel methods when they wish to check if two distributions are the same, rather than indirectly training a classifier.